Whole-Number Division: Up to Four Digits by One-Digit Divisors
Answer Key

Grades 4–5 • Fluency Practice

Quick Answers

1. 42	4. 400	7. 297	10. 545
2. 75	5. 405	8. 858	11. 5
3. 103	6. 129	9. 609	12. 66

Common wrong answers

6. *If they got 129.1: wrote the leftover as a decimal digit (129.1) instead of a remainder*

1. $168 \div 4$. Divide from the left. 1 hundred is smaller than 4, so start with 16 tens: $16 \div 4 = 4$ tens. Multiply and subtract: $16 - 16 = 0$. Bring down the 8: $8 \div 4 = 2$ ones. The quotient is $\boxed{42}$. Check: $42 \times 4 = 168$.

2. $375 \div 5$. 3 hundreds is smaller than 5, so start with 37 tens: $37 \div 5 = 7$ tens with 2 left ($5 \times 7 = 35$, $37 - 35 = 2$). Bring down the 5 to make 25: $25 \div 5 = 5$ ones. The quotient is $\boxed{75}$. Check: $75 \times 5 = 375$.

3. $618 \div 6$. $6 \div 6 = 1$ hundred; $6 - 6 = 0$. Bring down the 1: 1 is smaller than 6, so write **0** in the tens place and bring down the next digit. That makes 18: $18 \div 6 = 3$ ones. The quotient is $\boxed{103}$. The 0 is not optional — dropping it gives 13, which is far too small, since $13 \times 6 = 78$.

4. Estimating $2{,}835 \div 7$ with the compatible number 2,800: $28 \div 7 = 4$, so $2{,}800 \div 7 = 400$. Nadia's estimate is $\boxed{400}$. This also tells her the exact quotient will have 3 digits.

5. $2{,}835 \div 7$. 28 hundreds $\div 7 = 4$ hundreds; $28 - 28 = 0$. Bring down the 3: 3 is smaller than 7, so write **0** in the tens place. Bring down the 5 to make 35: $35 \div 7 = 5$ ones. The quotient is $\boxed{405}$, which is very close to the estimate of 400 from problem 4 — a good sign the work is right.

6. $517 \div 4$. $5 \div 4 = 1$ hundred with 1 left. Bring down the 1 to make 11: $11 \div 4 = 2$ tens with 3 left ($4 \times 2 = 8$, $11 - 8 = 3$). Bring down the 7 to make 37: $37 \div 4 = 9$ ones with 1 left ($4 \times 9 = 36$, $37 - 36 = 1$). Since 1 is smaller than 4, it stays as the remainder: $\boxed{129 \text{ R } 1}$. Check: $4 \times 129 + 1 = 516 + 1 = 517$.

Watch for: an answer written as 129.1. That is the remainder pasted on as a decimal digit, which is not what it means — the leftover 1 is 1 out of 4, not one tenth.

7. $891 \div 3$. $8 \div 3 = 2$ hundreds with 2 left. Bring down the 9 to make 29: $29 \div 3 = 9$ tens with 2 left. Bring down the 1 to make 21: $21 \div 3 = 7$ ones exactly. The quotient is $\boxed{297}$. Check: $297 \times 3 = 891$.

8. The missing number is the dividend. Division and multiplication undo each other, so if $\square \div 6 = 143$, then $\square = 143 \times 6$. Multiply: $143 \times 6 = 858$. The number is $\boxed{858}$. Check by dividing back: $858 \div 6 = 143$.

9. $4{,}872 \div 8$. 4 thousands is smaller than 8, so start with 48 hundreds: $48 \div 8 = 6$ hundreds; $48 - 48 = 0$. Bring down the 7: 7 is smaller than 8, so write **0** in the tens place. Bring down the 2 to make 72: $72 \div 8 = 9$ ones. The quotient is $\boxed{609}$. Check: $609 \times 8 = 4{,}872$.

10. Divide the pencils into groups of 6: $3{,}275 \div 6$. 32 hundreds $\div\ 6 = 5$ hundreds with 2 left ($6 \times 5 = 30$). Bring down the 7 to make 27: $27 \div 6 = 4$ tens with 3 left. Bring down the 5 to make 35: $35 \div 6 = 5$ ones with 5 left. So $3{,}275 \div 6 = 545$ R 5. The question asks only for *full* boxes, so the remainder is ignored here: $\boxed{545}$ full boxes.

11. Same division: $3{,}275 \div 6 = 545$ R 5. This time the question asks what is left after every full box is packed, and that is exactly the remainder: $\boxed{5}$ pencils. Check with the division equation: $6 \times 545 + 5 = 3{,}270 + 5 = 3{,}275$.

Teaching note: problems 10 and 11 are the same division. What changes is what the leftover means — that decision is the skill being practised.

12. Two divisions, one after the other.

Step 1 — tickets per class: $4{,}158 \div 7$. 41 hundreds $\div\ 7 = 5$ hundreds with 6 left ($7 \times 5 = 35$). Bring down the 5 to make 65: $65 \div 7 = 9$ tens with 2 left. Bring down the 8 to make 28: $28 \div 7 = 4$ ones. Each class gets 594 tickets.

Step 2 — tickets per student: $594 \div 9$. 5 is smaller than 9, so start with 59: $59 \div 9 = 6$ tens with 5 left ($9 \times 6 = 54$). Bring down the 4 to make 54: $54 \div 9 = 6$ ones. Each student receives $\boxed{66}$ tickets.

Estimate check: 4,158 is close to 4,200 and $4{,}200 \div 7 = 600$, so about 600 per class; $600 \div 9$ is between 60 and 70. The answer 66 fits.